

The Cl_7 Approach to the Standard Model

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ABSTRACT A recent geometric approach to the standard model in terms of the Clifford algebra Cl_7 is summarized. The complete gauge group of the standard model is shown to arise naturally and uniquely by considering all rotations in seven-dimensional space that (1) conserve the spacetime components of the particle and antiparticle currents and (2) do not couple the right-chiral neutrino. The spinor mediates a physical coupling of Poincaré and isotopic symmetries within the restrictions of the Coleman–Mandula theorem. The four extra spacelike dimensions in the model form a basis for the Higgs isodoublet field. The charge assignments of both the fundamental fermions and the Higgs boson are produced exactly.

Keywords: standard model, gauge groups, unification

34.1 Introduction

One of the most fundamental problems in particle physics is to explain the seemingly disparate gauge symmetries underlying the electromagnetic, weak and strong interactions. The prevailing wisdom is to embed the generators of these groups into some larger master group, but this leads to numerous theoretical problems. Larger groups contain extra gauge generators that should materialize as extra particle forces, and the present lack of any experimental evidence along these lines demands complicated theoretical remedies to explain their absence. Furthermore, invoking some master group does not address the question of the origin of gauge symmetries; one is still left with a large group inexplicably cast in some abstract space.

This paper demonstrates how the exact gauge symmetries

$$U(1)_Y \otimes SU(2)_L \otimes SU(3)_C$$

of the minimal standard model arise as the rotational symmetries of a reducible representation of the Poincaré group in a linear space with only four extra spacelike dimensions. Rather than embed the gauge groups into some master group,

AMS Subject Classification: 15A66, 81V22, 81T13.

we infix the Dirac algebra into the more mathematically uniform Clifford algebra $\mathcal{Cl}_7$. This is used to construct an algebraic approach that builds on a previous formulation [1] in geometric algebra of the Dirac theory. In our geometric model the gauge symmetries are not imposed but arise naturally from the algebra itself as unique symmetry groups of the current. Our approach of studying rotational symmetries in a higher-dimensional space may be viewed as an extension of the well-known association of spin with spatial rotations and the treatment of charge symmetry as a rotational symmetry in isospin space.

Section 2 summarizes the conventions adopted, as they vary throughout the Clifford algebra community. Section 3 develops the notion of spinors in $\mathcal{Cl}_7$. It shows how spinors representing all the fermions of a single generation can be combined into a single algebraic spinor, how the currents are calculated from such spinors, and how the contributions from individual fermions can be projected out. In section 4, we study the rotational symmetries of these spinors and show that they give exactly the gauge symmetries of the standard model with the correct weak hypercharge assignments. Section 5 shows how the four extra spatial dimensions and their transformation properties are precisely what is needed for the four components of a minimal Higgs field.

34.2 Algebraic foundations

In the real Clifford algebra $\mathcal{Cl}_7$, the unit vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_7$ are chosen to represent orthogonal spacelike directions in the tangent space of a seven-dimensional manifold. We further choose $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ to represent the three observed directions in physical space, and relegate $\mathbf{e}_4, \mathbf{e}_5, \mathbf{e}_6, \mathbf{e}_7$ to four hidden dimensions that are orthogonal to physical space. We will be primarily concerned with rotations in the tangent space, so the question of whether these extra dimensions are compact or infinite in extent will be tactfully avoided. It will later be shown how the particle current in these directions is equivalent to the Higgs field, so their presence is ultimately manifested as the mass of the particles.

The product of any number of vectors is completely determined by the anti-commutator $\mathbf{e}_j \mathbf{e}_k + \mathbf{e}_k \mathbf{e}_j = 2\delta_{jk}$, $j, k = 1, \dots, 7$. All elements of the algebra can be reduced to real linear combinations of $2^7 = 128$ basis forms, each one representing a geometric object. There are in general $\binom{7}{k}$ independent k -vectors in $\mathcal{Cl}_7$. Two basic antiautomorphic involutions are used. The reversion of $K \in \mathcal{Cl}_7$, denoted $K^\dagger$, reverses the order of appearance of all vector elements within K . Clifford conjugation, denoted by $\bar{K}$, both reverses the order and negates all vector elements of K . The algebra $\mathcal{Cl}_7$ is appealing in that the volume element of the algebra commutes with all elements and squares to -1 . It can therefore be associated identically with the unit imaginary

$$\mathbf{i} \equiv \mathbf{e}_1 \mathbf{e}_2 \mathbf{e}_3 \mathbf{e}_4 \mathbf{e}_5 \mathbf{e}_6 \mathbf{e}_7 \quad (2.1)$$

and used to reduce products of real vectors to elements of a complex space with